

Tanuj Pant

Full-Stack Engineer(TypeScript / Node.js / React / Next.js)

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Professional Summary

Senior Full-Stack Engineer with **7+ years of experience** building and scaling production applications across **React, Next.js, TypeScript, and Node.js**. Proven track record of improving performance and scalability, including **reducing 1 million-row data processing times by 86–95%**, building responsive and **virtualized interfaces for 10K+ records**, and automating operational workflows from ~3 weeks to 1–2 days. Experienced in designing distributed, event-driven systems using **PostgreSQL, Redis, Docker, and Kubernetes**, with hands-on work in microservices, **observability, and AI agent infrastructure**. Strong end-to-end ownership across system design, frontend, backend, infrastructure, performance optimization, and production reliability.

Key Skills

Languages: TypeScript, JavaScript, SQL

Frontend: React.js, Next.js, HTML5, CSS3, Tailwind CSS, TanStack Query

Backend: Node.js, Express.js, REST / RESTful APIs, WebSockets, Microservices, JWT, OAuth

Databases & Messaging: PostgreSQL, MySQL, Redis, Redis Streams, Pub/Sub

Cloud & DevOps: AWS, Docker, Kubernetes, CI/CD, GitHub Actions, S3, CloudFront

Testing: Jest, Vitest, Unit Testing, Integration Testing, End-to-End Testing

Architecture: Distributed Systems, Event-Driven Architecture, System Design, Idempotency

Observability: OpenTelemetry, Prometheus, Grafana, Tempo, Loki

Professional Experience

Full-Stack Engineer | Independent / Open-Source Projects, *Delhi NCR, India* | January 2026 - Present

Designed and built distributed systems spanning AI infrastructure and financial trading, including a **Kubernetes-based AI coding platform** with custom agent orchestration and an **event-driven perpetual futures matching engine**. Hands-on experience across system architecture, Node.js/TypeScript services, Redis Streams and pub/sub, PostgreSQL, WebSockets, Docker/Kubernetes, fault-tolerant processing, CI/CD, and end-to-end observability.

- Architected **distributed, event-driven systems** across an AI coding platform and perpetual futures trading engine using TypeScript, Node.js/Bun, Redis, PostgreSQL, Docker, Kubernetes, and WebSockets.
- Built a **custom AI agent harness** with LLM tool calling, **context management via compaction / summarization**, sub-agent **orchestration**, and isolated per-conversation Kubernetes workspaces with real-time application previews.
- Designed a high-performance **in-memory order matching engine with Redis Streams**, asynchronous persistence, **idempotent processing**, failure recovery, and independently deployable microservices.
- Implemented production infrastructure including workspace lifecycle management, AWS S3 backup/restore, **CI/CD**, distributed locking, and **observability with OpenTelemetry**, Prometheus, Tempo, Loki, and Grafana.
- Built the React/Next.js frontend and Node.js/Express.js APIs for conversation management, real-time WebSocket updates, AI-generated code execution, and live application previews.

Full Stack Developer, FreightExchange, Sydney Australia | November 2022 - January 2026

- Led the migration from a monolith to **REST-based microservices**, defining service boundaries and API contracts to improve platform scalability, maintainability, and independent service evolution.
- Architected **virtualized data tables for 10K+ records**, combining horizontal/vertical virtualization with externally hoisted state to preserve edits across row lifecycle changes; enabled responsive inline and bulk editing at scale.
- Automated carrier onboarding, rate management, and invoice reconciliation, transforming operationally heavy workflows into self-service configuration and mapping experiences and reducing manual processing from approx 3 weeks to 1-2 days.
- Implemented **JWT/OAuth** for authentication and authorization with unit and integration tests.
- **Integrated Stripe** billing to support the platform's transition to a SaaS subscription model and recurring revenue.

Software Engineer, Wootag, Singapore | April 2021 - November 2022

- Built React-based workflows for creating, configuring, previewing, and publishing interactive video ads across the campaign lifecycle.
- Built unit and end-to-end integration test coverage using Vitest and Jest for critical video-ad workflows.
- Integrated React authoring with a **lightweight Preact ad player via iframe and postMessage**, enabling real-time synchronization of ad configuration and metadata.
- Optimized publisher-side ad delivery around strict bundle-size constraints, using Preact and runtime isolation to maintain lightweight, performant ad experiences across 50+ publishers.

Senior Application Developer, Pinch / ACL Mobile, Noida, India | July 2018 - March 2021

- Built Vue.js & React.js applications end-to-end, owning UI architecture and integrations.
- Redesigned a **high-volume CSV ingestion pipeline processing 1-4M-row files**, optimizing the dominant ~80% new-record workload and **reducing end-to-end processing time by ~86-95%**.
- Re-architected background processing and database persistence for million-row uploads, substantially improving throughput, scalability, and reliability of large customer data imports.

Education

GNIIT - Computer Science

NIIT, Delhi India | GPA - 8.5 | April 2006 - March 2009

- 2+1 Year Structure: 2 years of classroom coding and software fundamentals followed by a mandatory 1-year paid industry internship.
- Core Technologies: Object-oriented programming, enterprise development environment
- Software Engineering Basics: Practical training in data structures, web development, relational database design, and the full Software Development Life Cycle (SDLC).
- Career Readiness: Dedicated modules for professional communication, business etiquette, interview prep, and live project experience.